Research Paper on-

Emergency Medical Service Dispatch & Telematics Platform

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Session:2026-27

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Research Paper Note

This paper presents the H8 Capability-Aware Emergency Medical Services (EMS) Dispatch & Telematics Platform described in the accompanying engineering and architecture dossier. The paper reorganizes the engineering specification into an academic research format, emphasizing the research problem, methodology, algorithmic contribution, system architecture, verification strategy, and simulation/benchmark findings.

The operational results reported in this paper are presented as simulation and engineering benchmark results from the project dossier. They should not be interpreted as evidence from a live clinical deployment or as measured patient-outcome improvements in a real ambulance service.

Abstract

Emergency Medical Services (EMS) dispatch is a time-critical decision problem in which the fastest vehicle is not necessarily the most clinically appropriate vehicle. Conventional closest-vehicle dispatch can produce capability mismatches, inefficient use of scarce Advanced Life Support (ALS) resources, reduced geographic coverage, and avoidable hospital offload delays. This paper presents H8, a capability-aware, distributed and event-driven EMS dispatch and telematics platform designed for metropolitan emergency response. The proposed system combines clinical capability matching, road-network travel-time estimation, crew fatigue, fleet coverage preservation, and telemetry freshness within a deterministic multi-factor dispatch scoring engine. The architecture integrates Java 21 and Spring Boot microservices with PostgreSQL/PostGIS, Redis geospatial tracking, Apache Kafka event streaming, Keycloak identity management, GraphHopper routing, and Progressive Web Applications (PWAs). A Maximum Expected Coverage Location Problem (MEXCLP) model is used to preserve backup coverage, while a hospital destination ranking component considers clinical capability, emergency-department capacity, diversion status, travel time, and offload delay. To support accountability, dispatch and clinical events are recorded in a SHA-256 hash-chained audit ledger. Verification combines property-based testing, deterministic Monte Carlo simulation with Common Random Numbers, concurrency testing, and performance benchmarking. According to the project dossier, simulations over 10,000 Jaipur metropolitan

incident scenarios reduced ALS clinical mismatch from 34.2% to 1.8%, reduced secondary intercept rate by 78%, and reduced hospital offload waiting time by 26.4 minutes per transport. Engineering benchmarks reported sub-15 ms P95 candidate-query-and-scoring latency and 12,400 Redis geospatial operations per second. These findings indicate that integrating clinical appropriateness with spatial, operational, and hospital-system constraints can provide a more robust dispatch decision framework than proximity-only assignment, while highlighting the need for prospective real-world validation before clinical deployment.

Keywords: Emergency Medical Services, Ambulance Dispatch, Capability-Aware Dispatch, Decision Support System, MEXCLP, Telematics, GraphHopper, Microservices, Redis Geo, Kafka, Hospital Diversion, Cryptographic Audit

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1. Introduction

Emergency Medical Services form a critical entry point into emergency healthcare. When a call is received, an EMS organization must make several decisions under severe time pressure: identify the urgency of the incident, determine the medical capability required, select an available ambulance, estimate travel time, preserve coverage for subsequent incidents, and coordinate with a suitable receiving hospital. These decisions are interdependent. A vehicle that is geographically close may be clinically unsuitable; a clinically suitable vehicle may be the only remaining resource in its zone; and a hospital that is geographically closest may be temporarily unable to receive the patient.

Research in EMS operations has therefore moved beyond simple static location models toward dynamic relocation, dispatching, routing, uncertainty management, and integrated emergency-care decision support. Reviews of the field identify dispatch, routing, ambulance location, relocation, hospital interaction, forecasting, and validation as interconnected operational problems [2–5]. The Maximum Expected Covering Location Problem (MEXCLP), introduced by Daskin, explicitly models the possibility that covering ambulances may be busy when a demand occurs [1]. More recent literature continues to investigate uncertainty, heterogeneous ambulance fleets, patient types, and joint location-dispatch decisions [6,7].

H8 is positioned within this research direction. Its central proposition is that ambulance dispatch should be treated as a capability-aware decision rather than a nearest-neighbour lookup. The platform combines clinical matching with travel-time, crew fatigue, coverage loss, and telemetry freshness factors. It further extends the dispatch decision into the transport phase by ranking hospitals according to clinical capability, capacity, diversion state, and expected offload delay.

The project is designed as an enterprise-oriented software platform for a metropolitan pilot in Jaipur, Rajasthan, with a documented target configuration of 14 active ambulance units and five Tier-1 receiving centers. The architecture uses event-driven microservices, spatial data infrastructure, real-time telemetry, and a cryptographically verifiable audit ledger. The research contribution is therefore not a single isolated algorithm; rather, it is the integration of several operational decision layers into one auditable technical architecture.

2. Problem Statement and Research Motivation

2.1 Existing Operational Challenge

A proximity-only ambulance dispatch policy generally ranks available vehicles by distance or estimated arrival time. Such a policy is attractive because it is simple, fast, and easy to explain. However, it implicitly assumes that all ambulances are clinically interchangeable and that the consequences of removing any one unit from the fleet are similar. These assumptions are weak in a heterogeneous emergency response system.

H8 identifies four major limitations in this conventional approach. First, capability-blind dispatch can send a Basic Life Support (BLS) unit to a case requiring Advanced Life Support (ALS) capability. Second, straight-line distance does not reliably represent road travel time in an urban environment. Third, selecting a vehicle without considering remaining fleet coverage can leave a demand zone with inadequate backup. Fourth, a hospital destination chosen without current capacity or diversion information can create avoidable offload delays.

Operational weakness	**Conventional approach**	**H8 response**
Vehicle selection	Closest available vehicle	Multi-factor capability-aware ranking
Travel estimation	Euclidean/straight-line distance	GraphHopper road-network ETA
Fleet readiness	Dispatch without coverage penalty	MEXCLP-based coverage preservation
Telemetry	Static/occasionally updated position	Redis Geo + freshness/dead-reckoning
Hospital destination	Nearest/known destination	Capability + capacity + diversion + delay
Audit trail	Conventional application logs	SHA-256 chained event ledger

Table 1. H8 operational problem-to-solution mapping.

2.2 Research Motivation

The research motivation is to construct a dispatch decision that balances two potentially conflicting goals: rapid response and clinically appropriate resource allocation. A purely distance-oriented objective can minimize travel time while increasing capability mismatch. Conversely, a capability-only rule could select an ALS unit from an unnecessarily distant location and degrade response time. H8 therefore formulates dispatch as a weighted decision problem in which clinical capability receives the highest empirical weight while travel time remains a major factor.

A second motivation is system-level resilience. An ambulance dispatch decision changes the future state of the entire fleet. If the last available unit in a geographic sector is dispatched, the next emergency in that sector may experience a much longer response. H8 incorporates a coverage-loss component and a redeployment service so that immediate dispatch decisions are considered together with future readiness.

A third motivation is accountability. EMS decisions can be operationally and legally sensitive. H8 therefore treats auditability as a first-class architectural concern. Dispatches, manual overrides, triage events, and status changes are recorded in a cryptographically linked ledger, allowing subsequent verification of event integrity.

3. Objectives and Research Questions

The primary objective of H8 is to develop and verify a capability-aware EMS dispatch and coordination platform that can select clinically appropriate ambulance resources while preserving operational coverage and coordinating hospital destinations.

Objective	**Research focus**
O1	Reduce clinically inappropriate ambulance assignments.
O2	Use road-network ETA rather than straight-line proximity for dispatch ranking.

O3	Preserve backup coverage while making immediate dispatch decisions.
O4	Incorporate telemetry freshness into dispatch confidence.
O5	Coordinate destination selection with hospital capability and diversion state.
O6	Provide secure, role-based and auditable operational workflows.
O7	Verify algorithmic invariants, concurrency behavior and performance.

Table 2. Research objectives and measurable focus.

3.1 Research Questions

1. Can a capability-aware scoring model reduce clinical mismatch compared with closest-vehicle dispatch under controlled simulation?
2. Can capability matching be combined with ETA without allowing clinically unsuitable but geographically closer units to dominate critical cases?
3. Can coverage-loss and redeployment mechanisms reduce the risk of leaving demand zones without adequate backup?
4. Can real-time telemetry and road-network routing be incorporated within operational latency targets?
5. Can dispatch decisions and manual overrides be made auditable without imposing prohibitive computational overhead?

3.2 Contribution of the Work

The contribution of this work is an integrated architecture in which the dispatch algorithm, spatial tracking, routing, fleet coverage, hospital coordination, security, event streaming, and verification strategy are designed as a single system. The paper also contributes a reproducible verification approach based on deterministic scoring, property-based tests, Common Random Numbers (CRN) for simulation comparison, and concurrency invariants.

4. Related Work and Research Gap

EMS optimization has a long history in operations research. Early work established mathematical models for ambulance location and coverage. Daskin's MEXCLP extended conventional coverage modeling by accounting for the probability that ambulances capable of covering a demand point may be busy when an emergency occurs [1]. This is directly relevant to H8 because the platform seeks to preserve backup coverage rather than treating every available unit as permanently available.

Brotcorne, Laporte, and Semet reviewed ambulance location and relocation models and showed the progression from deterministic planning models toward probabilistic and dynamic models that account for ambulance availability and repeated relocation [2]. Bélanger, Ruiz, and Soriano later reviewed optimization models for location, relocation, and dispatching, emphasizing uncertainty, dynamism, social objectives, and patient outcomes [3]. Aringhieri and colleagues provided a broader EMS literature review covering location, relocation, dispatching, routing, hospital interactions, evaluation, forecasting, and workforce considerations [4].

Simulation is also important because EMS systems are stochastic and highly dynamic. A review of simulation models in EMS reported extensive use of simulation for evaluating ambulance location, deployment, and redeployment strategies [5]. Recent work continues to address multi-tier ambulance systems, patient types, time-dependent demand, and travel-time variation [6]. A 2026 integrated decision-support study further demonstrates the continuing movement toward models that jointly consider dispatch, patient-to-hospital allocation, and ambulance relocation [7].

4.1 Research Gap Addressed by H8

The project dossier identifies a practical gap between isolated optimization models and an operationally integrated software platform. Many research models focus on one decision layer—such as location, relocation, or dispatch—whereas a deployable EMS system must coordinate these decisions with live telemetry, hospital capacity, security, event delivery, and auditability. H8 addresses this systems-engineering gap by implementing the algorithmic core within a microservice architecture and by treating dispatch, coverage, routing, destination selection, and audit as interacting components.

Another gap is the explicit inclusion of clinical capability as a dominant dispatch factor. H8 does not claim that its particular weight values are clinically optimal in all jurisdictions. Instead, it provides a deterministic and tunable mechanism in which clinical mismatch can be represented mathematically and tested through invariants. This makes the approach suitable for later calibration with local EMS policy and clinical governance.

Research direction	**Prior literature emphasis**	**H8 implementation emphasis**
Coverage	Facility/ambulance coverage and busy probability	MEXCLP + real-time coverage-loss penalty
Dispatch	Response time, location, dispatch rules	Clinical capability + ETA + fatigue + coverage + freshness
Relocation	Dynamic repositioning	Redeployment recommendations after coverage degradation
Routing	Travel time and network decisions	GraphHopper road graph + fallback
Hospital interaction	Destination and emergency-care pathway	Capacity, diversion, capability and offload delay

Validation	Mathematical models / simulation	Property tests + CRN simulation + concurrency + benchmarks

Table 3. Positioning H8 within related EMS research.

5. Proposed H8 System

H8 is designed as a distributed, event-driven EMS decision-support platform. Its primary actors are emergency dispatchers, ambulance crews, hospital staff, and auditors. A caller initiates an incident; triage information is converted into a deterministic severity and clinical-need representation; the dispatch service retrieves suitable available units; the routing service supplies road-based ETAs; the scoring engine ranks candidates; and an atomic reservation prevents two simultaneous dispatches from claiming the same ambulance.

After assignment, the tracking service receives high-frequency telemetry and stores the latest location in Redis geospatial structures. During transport, the destination-ranking service considers receiving-hospital capability and real-time operational state. Hospital staff can update capacity and diversion state, while the crew receives pre-arrival information. All significant state changes are propagated through Kafka and recorded for audit.

5.1 Incident Lifecycle

The documented incident lifecycle is represented as REPORTED → TRIAGED → DISPATCHED → ON_SCENE → TRANSPORTING → RESOLVED, with cancellation supported as an alternative terminal state. Illegal state transitions return HTTP 409 Conflict. This explicit lifecycle prevents ambiguous operational state and supports event-driven synchronization between services.

5.2 Clinical Representation

H8 represents incident severity using deterministic categories including CRITICAL, URGENT, STANDARD, and NON_EMERGENCY. Clinical need categories include CARDIAC, TRAUMA, STROKE, PEDIATRIC, RESPIRATORY, and GENERAL. A boolean requirement for ALS

capability is derived from triage information. The purpose is not to replace professional clinical judgment but to encode operationally relevant requirements for resource selection.

5.3 User-facing Workflows

- Dispatcher command center: incident queue, live fleet map, candidate ranking, coverage indicators and audited manual override.
- Paramedic PWA: mission status, route guidance, vital signs, destination ranking and pre-arrival transmission.
- Hospital ED dashboard: bed/capacity information, incoming patient alerts and diversion control.
- Audit interface: read-only access for authorized auditors to cryptographically verifiable dispatch history.

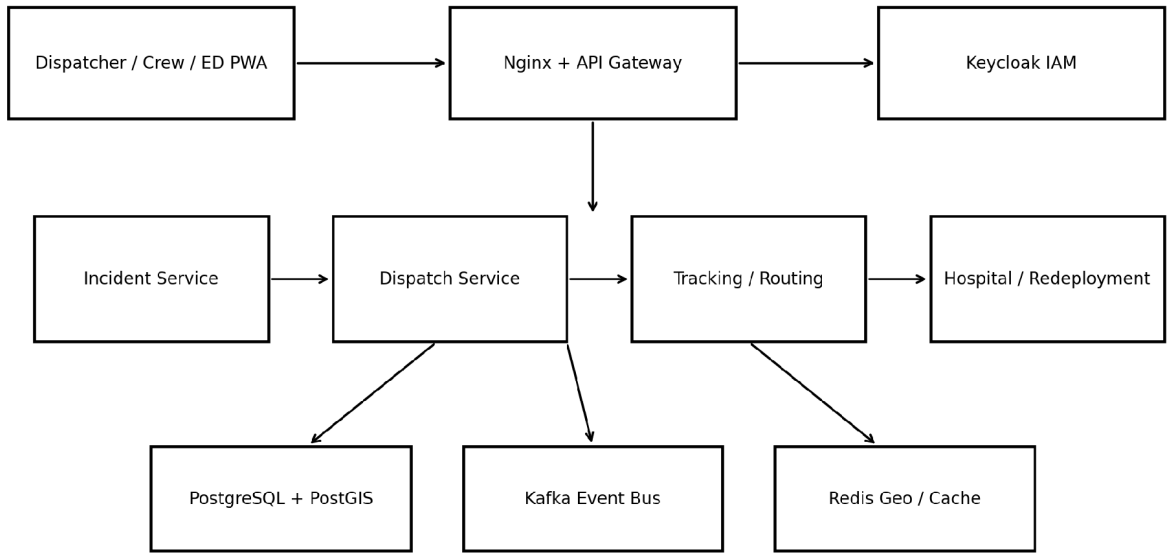


Figure 1. High-level H8 architecture derived from the project architecture dossier.

6. System Architecture and Design

The platform uses a distributed microservices architecture based on Domain-Driven Design principles. The algorithmic core follows Hexagonal (Ports and Adapters) architecture so that the core scoring and coverage logic remains independent of Spring, JPA, Kafka, and Redis. This separation is particularly important for verification because the same pure Java scoring components can be reused by the offline simulation runner.

6.1 Service Decomposition

Service	Primary responsibility	Key infrastructure
API Gateway	JWT validation, routing, CORS and external entry point	Spring Cloud Gateway
Incident Service	Call intake, triage, caller hashing and incident state	PostgreSQL
Dispatch Service	Candidate retrieval, scoring and atomic reservation	PostgreSQL + Redis
Tracking Service	GPS ingestion, spatial index, freshness and extrapolation	Redis Geo
Routing Service	Road routes, ETAs and travel-time matrix	GraphHopper + OSM graph
Hospital Service	Capacity, diversion and destination ranking	PostgreSQL
Redeployment Service	Coverage analysis and move-up recommendations	Coverage model + spatial data
Audit Service	Append-only cryptographic ledger and verification	PostgreSQL + SHA-256

Table 4. H8 service architecture.

6.2 Synchronous and Asynchronous Communication

Latency-sensitive query paths use synchronous REST communication. For example, Dispatch Service requests a road travel-time matrix from Routing Service during candidate ranking. If routing exceeds the documented timeout threshold, the architecture specifies a fallback to Haversine distance with an urban tortuosity factor. State-changing operations use the Transactional Outbox Pattern: a service commits the domain mutation and its corresponding event record in one database transaction, after which an outbox relay publishes the event to Kafka.

Kafka consumers are designed for at-least-once delivery and idempotent processing. Consumers check a processed-events record before applying state changes. This pattern reduces the risk that duplicated messages produce duplicated assignments or inconsistent state.

6.3 Concurrency and Atomicity

Ambulance reservation is implemented as an atomic conditional update from AVAILABLE to ASSIGNED. If two dispatch requests attempt to reserve the same unit concurrently, only one transaction can successfully update the available row. The remaining request receives an unavailable/conflict response and must re-rank candidates. This is a critical invariant because duplicate reservation of one ambulance would directly compromise operational correctness.

7. Dispatch Optimization Methodology

The H8 dispatch engine evaluates every eligible ambulance using five normalized components. The score is designed so that higher values represent more desirable candidates.

The overall score is:

$$\mathbf{S(u,i) = \alpha\, fETA(u,i) + \beta\, fcap(u,i) + \gamma\, ffatigue(u) + \delta\, fcov(u) + \varepsilon\, fstale(u)}$$

The weights documented in the project are $\alpha = 0.35$ for proximity/ETA, $\beta = 0.45$ for clinical capability, $\gamma = 0.08$ for crew fatigue, $\delta = 0.07$ for coverage loss, and $\varepsilon = 0.05$ for telemetry staleness. The weights sum to 1.0.

Component	Weight	Purpose
Proximity / ETA	0.35	Favor fast road-network arrival.
Capability matching	0.45	Prioritize clinically appropriate equipment and care level.
Crew fatigue	0.08	Reduce operational risk associated with extended workload.
Coverage loss	0.07	Avoid stripping the last backup resource from an area.
Telemetry staleness	0.05	Reduce confidence in outdated vehicle positions.

Table 5. Dispatch scoring weights documented in H8.

7.1 ETA Component

The ETA component is normalized using a maximum reference ETA of 1,800 seconds. The dossier specifies a monotonically decreasing formulation: fETA = max(0, 1 − ETA/ETAmax). Thus, a lower ETA yields a higher contribution to the final score. Road-network ETA is preferred over straight-line distance because urban roads, turns, and route geometry can make Euclidean proximity misleading.

7.2 Capability Component

The capability component is defined as fcap = 1 − P mismatch, where Pmismatch is obtained from a clinical mismatch matrix. For incidents requiring ALS capability, a BLS unit receives a documented mandatory penalty of 0.55. More severe or clinically specific combinations can receive additional penalties according to the matrix. The design deliberately gives capability the highest weight,

reflecting the project hypothesis that clinical suitability should dominate minor proximity advantages in high-acuity cases.

7.3 Fatigue, Coverage and Telemetry

Crew fatigue contributes a smaller but non-zero factor. Coverage loss is computed from the remaining available units in an approximately 8 km neighborhood and a target backup density. If dispatching a unit leaves its zone without sufficient backup, its score is reduced. Telemetry freshness is modeled exponentially: $f_{stale} = \exp(-\Delta t/T_{decay})$, with $T_{decay} = 30$ seconds. The documented thresholds treat telemetry older than 60 seconds as stale.

7.4 Algorithmic Invariants

6. If two otherwise identical units differ only in travel distance, the closer unit must have an equal or higher score.
7. For a critical cardiac incident, a clinically appropriate ALS unit at a moderate distance must rank above an inappropriate BLS unit that is substantially closer, when the capability penalty is intended to dominate.
8. For concurrent dispatch attempts targeting the same unit, exactly one assignment succeeds and the remaining attempts fail with a conflict/unavailable outcome.
9. Stale telemetry must never be treated as equally trustworthy as fresh telemetry.

These invariants are important because they test the intended behavior of the scoring model rather than only checking whether code executes without exceptions.

8. Coverage Optimization and Hospital Destination Selection

8.1 MEXCLP-based Coverage Preservation

H8 applies the Maximum Expected Coverage Location Problem framework to metropolitan readiness. Let demand nodes be represented by I, candidate base stations by J, fleet size by p, demand weight at node i by d_i, and ambulance busy probability by q. The project uses a mean service time of approximately 45 minutes in its documented formulation. The model estimates

expected coverage while accounting for the possibility that one or more nearby ambulances are busy.

The project formulation uses an 8 km neighborhood as a response-coverage threshold and calculates expected coverage across backup levels. In operational terms, the model prevents a local dispatch decision from being evaluated only against the current incident. It also asks what emergency readiness remains after the unit leaves.

8.2 Dynamic Redeployment

When a dispatch reduces coverage in a suburban or otherwise vulnerable quadrant, the redeployment service can recommend move-up locations for idle ambulances. The purpose is to reduce secondary response delays and maintain a minimum level of metropolitan readiness. This connects immediate dispatch with future demand protection.

8.3 Hospital Destination Ranking

The destination-ranking layer extends the decision chain beyond ambulance-to-incident assignment. Hospitals are evaluated using clinical capability alignment, travel time, diversion status, and offload delay. Examples of capability alignment include cardiac patients requiring cath-lab support and stroke patients requiring suitable neurological capability. Hospitals can declare diversion when emergency-department capacity is exhausted or a critical diagnostic capability is unavailable.

During transport, the hospital receives a real-time pre-arrival alert containing relevant operational information such as patient acuity, ETA, and vital signs. This creates a closed information loop between the ambulance and receiving facility.

9. Security, Privacy and Auditability

Because EMS platforms handle operational and potentially sensitive patient information, H8 incorporates security at the architecture level rather than treating it as a later add-on. Authentication and role-based authorization are provided through Keycloak using OAuth 2.0/OpenID Connect tokens. Four documented roles are DISPATCHER, CREW, HOSPITAL_STAFF, and AUDITOR.

Control	H8 mechanism	Purpose
Authentication	Keycloak / OAuth2 / OIDC	Verify user identity.
Authorization	Role-based access control	Restrict operational actions by role.
Transport security	TLS 1.3	Protect data in transit.
Caller privacy	Salted/HMAC-SHA-256 phone digest	Avoid storing plaintext caller phone numbers.
Audit integrity	SHA-256 hash chain	Detect modification of historical events.
Secrets	Environment variables / secret management	Prevent credentials in source control.

Table 6. Security and privacy controls.

9.1 Cryptographic Audit Ledger

Each audit block contains a sequence index, previous hash, event identifier, canonical event payload, timestamp, and current block hash. The documented computation is:

$$\mathbf{H_i = SHA\text{-}256(H_{i-1} \,||\, EventUUID_i \,||\, PayloadJSON_i \,||\, TimestampEpoch_i)}$$

The genesis hash is a 64-character zero value. During verification, the system recalculates each block and checks both the predecessor pointer and the block hash. If an earlier payload is modified, the mismatch propagates into subsequent blocks. This provides tamper evidence for the application-level audit history.

9.2 Privacy-by-Design Considerations

The project specifies that caller phone numbers are transformed into a salted digest before database insertion and are not persisted or logged in plaintext. This design reduces direct exposure of caller identifiers. However, privacy protection in a real deployment would also require a formal

data-retention policy, access review, encryption-at-rest strategy, incident response procedures, legal review, and jurisdiction-specific compliance validation. The paper therefore treats the documented controls as an architectural baseline rather than a claim of complete regulatory compliance.

10. Implementation and Technology Stack

The documented implementation uses Java 21 LTS and Spring Boot 3 for backend services. PostgreSQL 15 with PostGIS 3.4 provides transactional and spatial persistence. Redis 7 provides low-latency geospatial indexing for active ambulance positions. Apache Kafka 3.7 provides event streaming. Keycloak 25 provides identity and access management. GraphHopper provides road-network routing and travel-time matrix computation. The frontend is delivered as a Progressive Web Application.

Layer	**Technology**	**Role**
Backend	Java 21 LTS / Spring Boot 3	Core application services and domain logic
Spatial database	PostgreSQL 15 + PostGIS 3.4	Persistent operational and geospatial data
Event streaming	Apache Kafka 3.7	Asynchronous domain events
Real-time location	Redis 7 Geo	Low-latency fleet position indexing
Identity	Keycloak 25	Authentication and RBAC
Routing	GraphHopper	Road-network route and ETA calculation
Frontend	PWA	Dispatcher, crew and hospital interfaces
Deployment	Docker / Ubuntu 24.04	Containerized operational environment

Table 7. Technology stack documented for H8.

10.1 Frontend Architecture

The dispatcher interface provides a high-density operational map, candidate rankings, incident queue, manual override and coverage indicators. The crew interface supports mission lifecycle updates, route guidance and clinical vital entry. The emergency-department interface provides pre-arrival alerts, capacity visibility and diversion control. The architecture also documents an offline/demo bridge using browser communication mechanisms and a pure-JavaScript scoring path for demonstration scenarios.

10.2 Deployment Model

The dossier defines development/test infrastructure at approximately 4 vCPU and 8 GB RAM, metropolitan production around 8 vCPU and 16 GB RAM for 14–100 units, and a multi-region Kubernetes approach for larger national-scale deployments. Health probes are exposed through Spring Boot Actuator, and operational procedures include database backup, restoration, rolling updates, and monitoring.

11. Verification and Experimental Methodology

Verification is central to H8 because dispatch software must be correct not only under normal input but also under concurrency, stale telemetry, routing failure, and edge-case clinical combinations. The project uses a three-tier verification strategy: property-based testing, deterministic Monte Carlo simulation, and integration/concurrency verification.

11.1 Property-based Testing

The project uses jqwik-style property-based testing to generate diverse combinations of units, locations, clinical needs, severities, fatigue states, coverage conditions, and telemetry ages. Rather than checking only fixed examples, the tests assert general properties of the scoring function and reservation logic. This is particularly useful for identifying edge cases in weighted optimization and state transitions.

11.2 Deterministic Monte Carlo Simulation

The simulation compares two policies: a naive closest-vehicle policy and the H8 capability-aware policy. Common Random Numbers (CRN) are used so that both policies receive the same incident streams and stochastic inputs. This reduces variance caused by random sampling and makes observed differences more attributable to the dispatch policy.

11.3 Integration and Concurrency Testing

Integration tests verify service-to-service interactions, event publication, idempotent consumers, routing fallback, database transactions and audit-chain behavior. Concurrency tests target the atomic unit reservation invariant. A successful dispatch must reserve one available unit exactly once, while competing requests must fail safely rather than creating duplicate assignments.

Verification tier	Primary question	Expected evidence
Property-based tests	Does the algorithm preserve mathematical invariants?	Pass/fail properties over generated cases
CRN Monte Carlo	Does H8 outperform closest dispatch under identical scenarios?	Matched-policy performance comparison
Integration tests	Do services preserve intended state and event contracts?	End-to-end transaction/event results
Concurrency tests	Can one ambulance be assigned twice?	Single successful reservation + conflicts
Performance benchmarks	Does the system meet latency/throughput goals?	Measured execution times and throughput

Table 8. Verification strategy.

12. Results and Performance Analysis

The results in this section are reported from the project dossier and are therefore treated as engineering benchmark and simulation evidence. They demonstrate the behavior of the implemented design under the documented test conditions; they do not establish clinical effectiveness in live patient care.

12.1 Software Performance Benchmarks

Benchmark scenario	Metric	Reported result
DispatchScorer (100 units)	Computation latency	0.84 ms mean
Candidate query + scorer	Database + scoring	14.2 ms P95
Telemetry ingestion	Redis GEO throughput	12,400 operations/sec
GraphHopper 14×1 matrix	Route calculation	28.5 ms P95
SHA-256 audit append	Block operation	1.12 ms/block
Audit verification	10,000 blocks	380 ms total

Table 9. Engineering performance benchmarks reported in the project dossier.

The reported 0.84 ms mean scoring time indicates that the mathematical ranking engine itself is computationally lightweight relative to the surrounding distributed-system operations. The 14.2 ms P95 full candidate-query-and-scoring result suggests that database retrieval plus ranking can remain within a low-latency operational budget for the tested configuration. GraphHopper routing is more expensive than pure scoring but remains below the documented 300 ms fallback threshold under the benchmark scenario.

12.2 Operational Simulation Results

Metric	Naive closest dispatch	H8 simulation result	Change
ALS clinical mismatch	34.2%	1.8%	−32.4 percentage points
Secondary intercept rate	Baseline	78% reduction	−78%
Cardiac secondary-response time	Baseline	11.4 min average saving	Improved
Hospital offload wait	Baseline	26.4 min reduction/transport	Improved

Table 10. Simulation-based operational comparison over 10,000 Jaipur metropolitan scenarios.

The most prominent reported result is the reduction in ALS clinical mismatch from 34.2% to 1.8%. This result is consistent with the algorithm's design because capability matching carries the highest weight and clinically inappropriate BLS candidates receive a substantial penalty for high-acuity cases. The 78% reduction in secondary intercept rate is consistent with the inclusion of coverage loss and redeployment logic, although the magnitude of the effect depends on the simulation assumptions and fleet configuration.

The reported 26.4-minute reduction in hospital offload waiting time illustrates the value of extending decision support into the destination-selection stage. Instead of selecting a hospital based only on distance, the system can avoid facilities under diversion or with unsuitable capacity/capability. Again, this is a simulation result and should be validated against real historical or prospective hospital-flow data before operational claims are made.

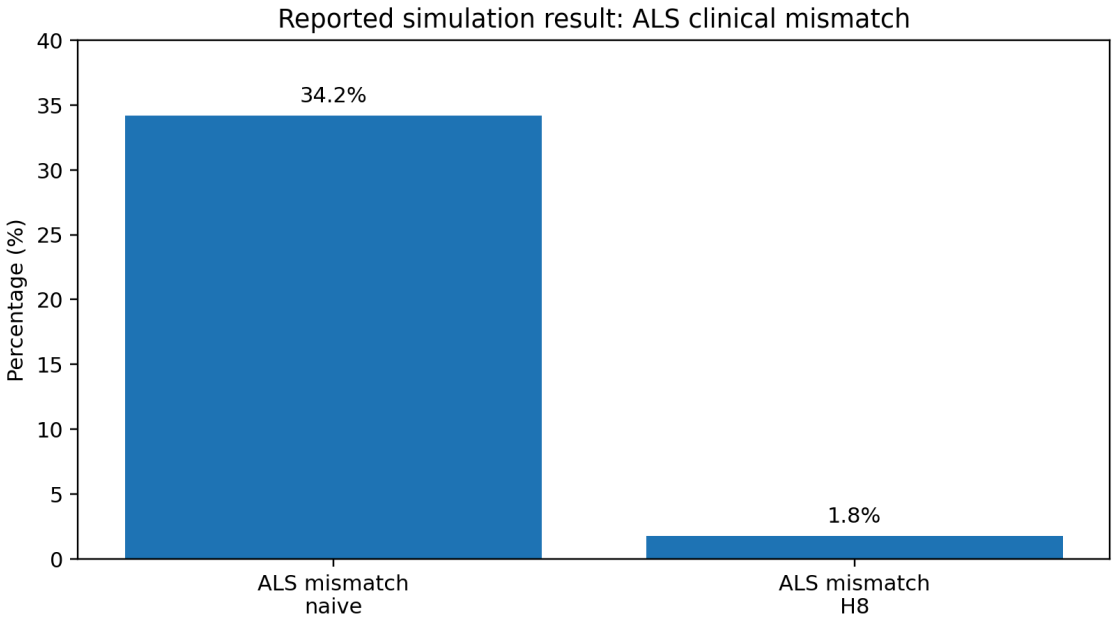


Figure 4. Reported simulation comparison for ALS clinical mismatch. Values are from the project dossier.

13. Discussion

The H8 results support the central hypothesis that ambulance dispatch quality cannot be represented by distance alone. The dispatch score explicitly gives capability matching a 0.45 weight, higher than the 0.35 ETA weight. This is an intentional design choice: a marginally faster but clinically unsuitable vehicle should not automatically defeat a clinically appropriate alternative in a high-acuity emergency.

At the same time, the system does not ignore response time. The ETA factor remains the second-largest weight and is calculated from road-network travel time. This creates a balanced optimization objective in which clinical appropriateness and rapid arrival are considered together. The architecture further avoids over-reliance on one data source by incorporating telemetry freshness and routing fallback.

The coverage component represents a second-order operational effect. Dispatching one unit solves the current incident but may create risk for the next incident. MEXCLP-based reasoning and redeployment recommendations address this system-level coupling. This is aligned with the broader

EMS literature, which emphasizes that ambulance location, relocation and dispatch decisions are interconnected rather than independent [2–4].

The hospital component is another important systems-level contribution. Emergency response does not end at the incident scene. A patient may experience delay if an ambulance reaches a hospital that cannot receive the patient efficiently. By incorporating diversion state, clinical capability and offload delay into destination ranking, H8 extends the optimization boundary from ambulance dispatch to the receiving-facility interface.

13.1 Practical Significance

- Dispatchers receive an explainable ranking instead of a black-box vehicle choice.
- Manual overrides remain possible but require a documented justification.
- Ambulance positions are treated as time-sensitive data rather than permanent coordinates.
- Hospital capacity and diversion are integrated into transport decisions.
- Operational events remain traceable through a cryptographically verifiable ledger.
- The pure-Java domain layer can be tested independently of infrastructure components.

13.2 Research Significance

From a research perspective, H8 demonstrates how classical EMS operations-research concepts can be embedded into a modern software architecture. MEXCLP provides the conceptual basis for coverage preservation; deterministic multi-factor scoring provides the dispatch mechanism; simulation provides policy comparison; and microservices provide a deployable systems boundary. The combination is particularly relevant to applied computer science and software engineering because it connects mathematical decision models with implementation, security, testing, and operational observability.

14. Limitations and Threats to Validity

Several limitations must be acknowledged before interpreting the results as evidence of real-world clinical improvement.

10. The reported 10,000-incident evaluation is simulation-based. It does not constitute a prospective trial or a live EMS deployment.
11. The weight values in the dispatch score are documented engineering choices and have not been established as universally optimal across different EMS jurisdictions.
12. Clinical capability categories and mismatch penalties require validation by qualified EMS clinicians and local medical protocols.
13. Simulation outcomes depend on the assumed demand distribution, service duration, fleet size, hospital capacities, travel times and other generated parameters.
14. Real-world traffic, road closures, driver behavior, weather, hospital congestion and communication outages may differ substantially from model assumptions.
15. The audit hash chain provides tamper evidence but is not by itself a complete compliance framework.
16. Privacy and security requirements for production healthcare systems would require jurisdiction-specific legal and governance review.
17. The architecture targets a metropolitan pilot and therefore requires additional validation before national or multi-region scaling.

14.1 Threats to External Validity

Results obtained from Jaipur-specific simulation scenarios may not generalize directly to another city with different road topology, ambulance mix, hospital distribution, demand patterns, traffic conditions or clinical protocols. The project should therefore be considered a platform and methodology whose parameters can be calibrated rather than a universally fixed dispatch policy.

14.2 Threats to Internal Validity

The CRN approach strengthens internal comparison because both policies receive matched random inputs. Nevertheless, internal validity depends on correct simulator implementation and faithful representation of the baseline policy. Reusing the same pure domain implementation for production and simulation reduces behavioral divergence, but independent validation of the simulator remains desirable.

15. Future Scope

The H8 architecture provides several directions for future research and development.

- Clinical calibration: involve EMS physicians and paramedics to calibrate capability penalties, severity rules and resource priorities.
- Traffic-aware routing: incorporate live traffic and road-closure feeds rather than relying only on a static road graph.
- Predictive demand modeling: forecast near-term incident demand by time, location and historical patterns to improve redeployment.
- Learning-assisted scoring: evaluate machine-learning or reinforcement-learning methods under strict safety constraints while retaining deterministic clinical guardrails.
- Outcome-aware optimization: connect dispatch decisions with validated clinical outcomes rather than using response-time proxies alone.
- Multi-hospital coordination: incorporate ICU, trauma, cath-lab, stroke and pediatric capacity at finer granularity.
- National-scale deployment: evaluate multi-region Kubernetes, fault isolation and cross-region event replication.
- Human factors research: measure dispatcher workload, trust, override behavior and explanation quality.

- Prospective evaluation: conduct controlled pilot studies with historical replay followed by shadow-mode and carefully governed live deployment.

A particularly important future direction is outcome-based evaluation. Recent reviews note that EMS relocation and dispatch research has often emphasized mathematical or response-time objectives while evidence about patient outcomes and real-world system effects remains limited [8]. H8 should therefore evolve from simulation validation toward clinically governed prospective evaluation.

16. Conclusion

This paper presented H8, a capability-aware Emergency Medical Services dispatch and telematics platform designed to address limitations of proximity-only ambulance dispatch. The system reframes ambulance selection as a multi-factor decision involving clinical capability, road-network ETA, crew fatigue, remaining fleet coverage, and telemetry freshness. It then extends the decision process into dynamic redeployment and hospital destination selection.

The architecture combines microservices, event-driven messaging, spatial databases, Redis geospatial tracking, GraphHopper routing, Keycloak security, Progressive Web Applications, and a SHA-256 hash-chained audit ledger. The algorithmic core is separated through Hexagonal Architecture so that scoring and coverage logic can be independently tested and reused in simulation. The verification strategy combines property-based testing, Common Random Numbers, integration testing, concurrency verification and performance benchmarking.

According to the engineering dossier, simulations across 10,000 Jaipur metropolitan scenarios reduced ALS clinical mismatch from 34.2% to 1.8%, while secondary intercept rate fell by 78% and hospital offload waiting time was reduced by 26.4 minutes per transport. Engineering benchmarks also reported low-latency scoring and high-throughput telemetry processing. These findings support the feasibility of the architecture and the central research hypothesis, but they should be interpreted as simulation and engineering evidence rather than clinical proof.

The principal conclusion is that effective EMS dispatch should be viewed as a coordinated decision-support problem rather than a nearest-vehicle search. By combining clinical appropriateness, spatial optimization, fleet resilience, hospital readiness, security, and auditability in

one platform, H8 provides a strong foundation for future controlled evaluation and potential real-world deployment.

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Appendix A. Key Mathematical Formulations

Dispatch objective:

$$S(u,i) = 0.35\, f_{ETA} + 0.45\, f_{cap} + 0.08\, f_{fatigue} + 0.07\, f_{cov} + 0.05\, f_{stale}$$

Telemetry freshness:

$$f_{stale}(u) = \exp(-\Delta t_{last_fix} / 30)$$

Audit chain:

$$H_i = \text{SHA-256}(H_{i-1} \,||\, \text{EventUUID}_i \,||\, \text{PayloadJSON}_i \,||\, \text{TimestampEpoch}_i)$$

The mathematical formulations above are reproduced in condensed academic form from the project engineering dossier. They are included to make the research methodology self-contained.

Appendix B. Suggested Evaluation Protocol for Future College Demonstration

18. Replay a fixed historical or synthetic incident stream through both the closest-vehicle baseline and H8.
19. Use identical random seeds or CRN streams for matched comparisons.
20. Measure clinical mismatch, response ETA, coverage loss, secondary intercepts and hospital offload delay.
21. Run concurrency tests with multiple simultaneous dispatch requests against the same ambulance.
22. Inject routing-service failures and stale telemetry to verify graceful degradation.
23. Verify audit-chain integrity after normal operations and after controlled tampering tests.
24. Report confidence intervals and sensitivity analysis for scoring weights before drawing general conclusions.